Conducting system of the heart

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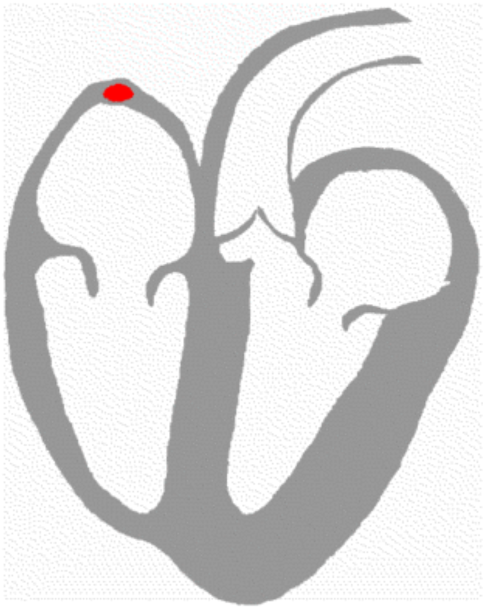
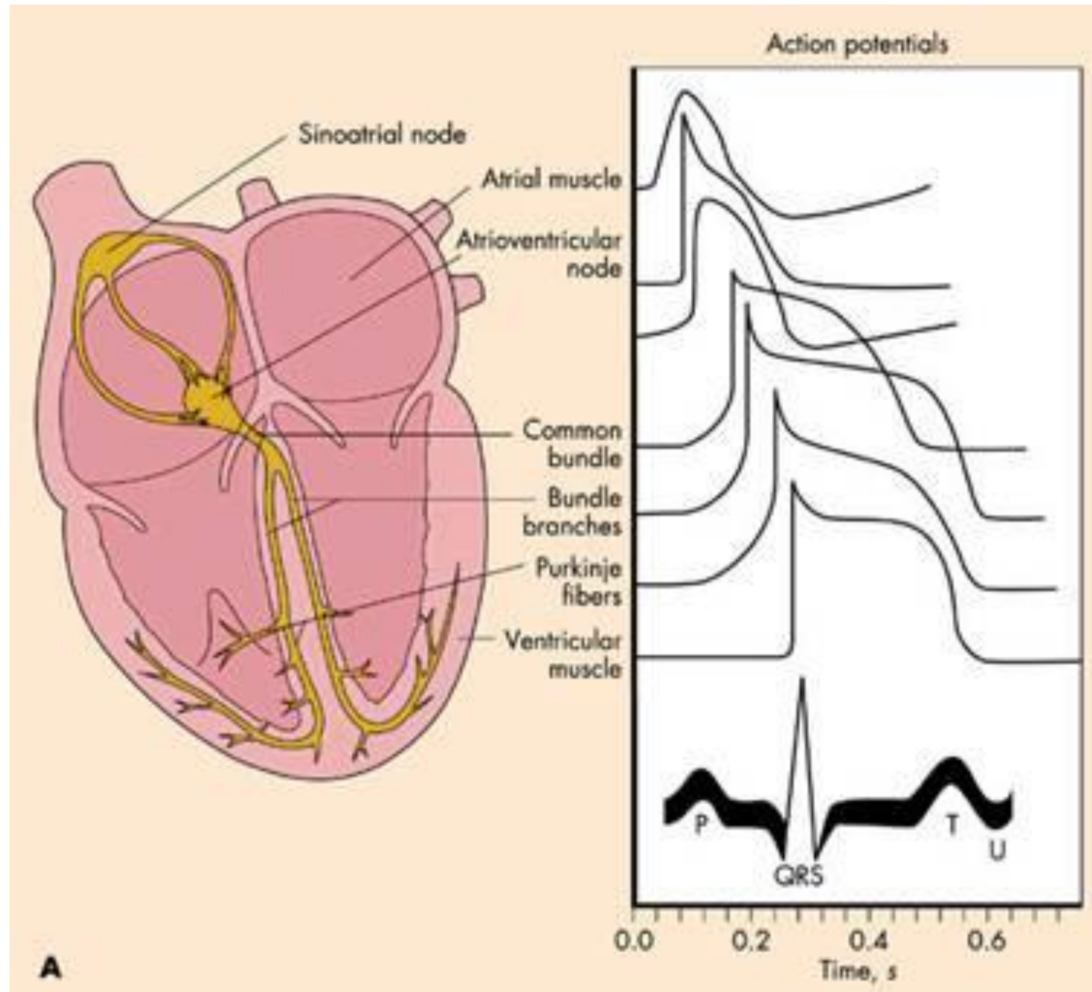
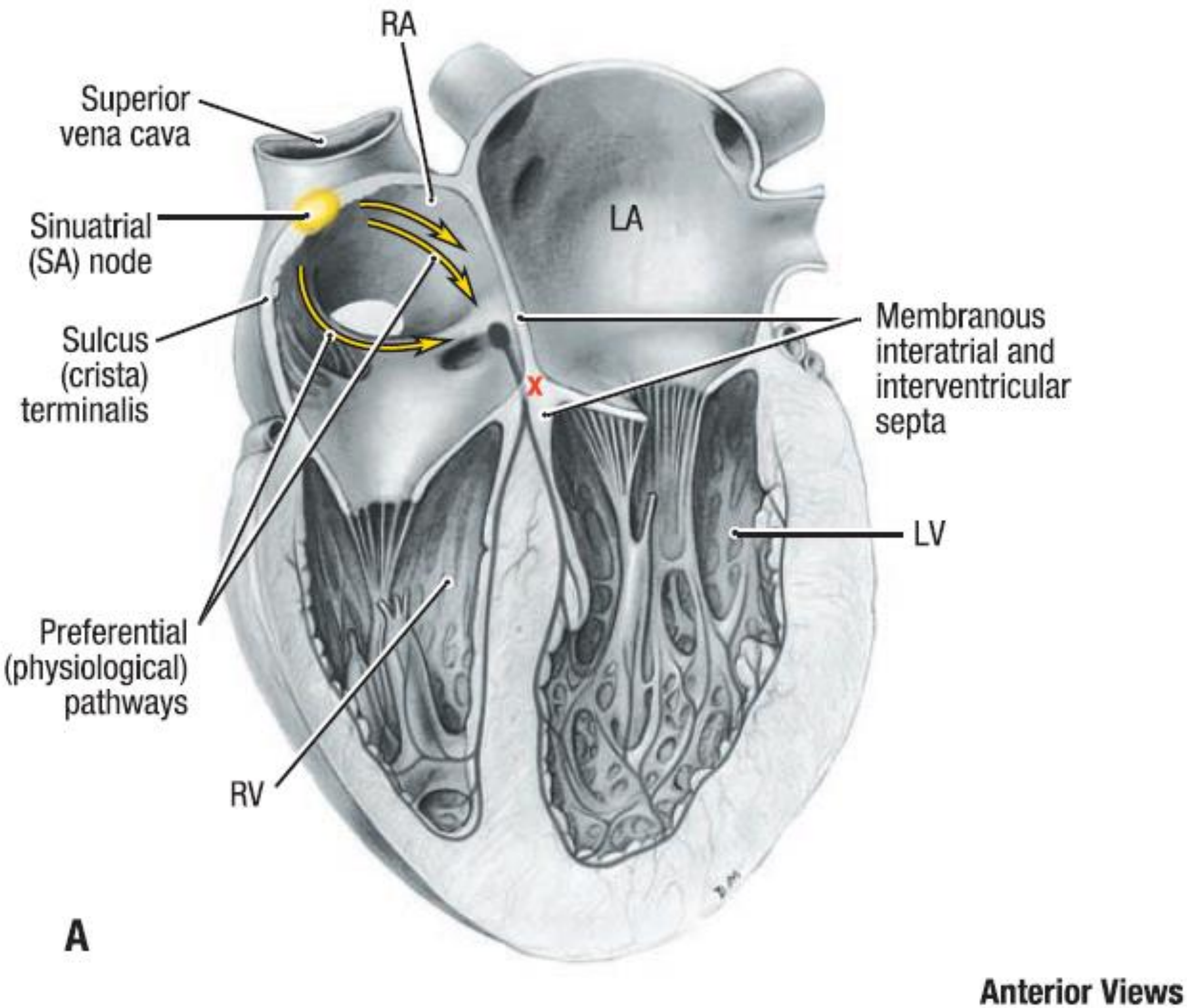


Electrocardiogram



Conducting system of the heart

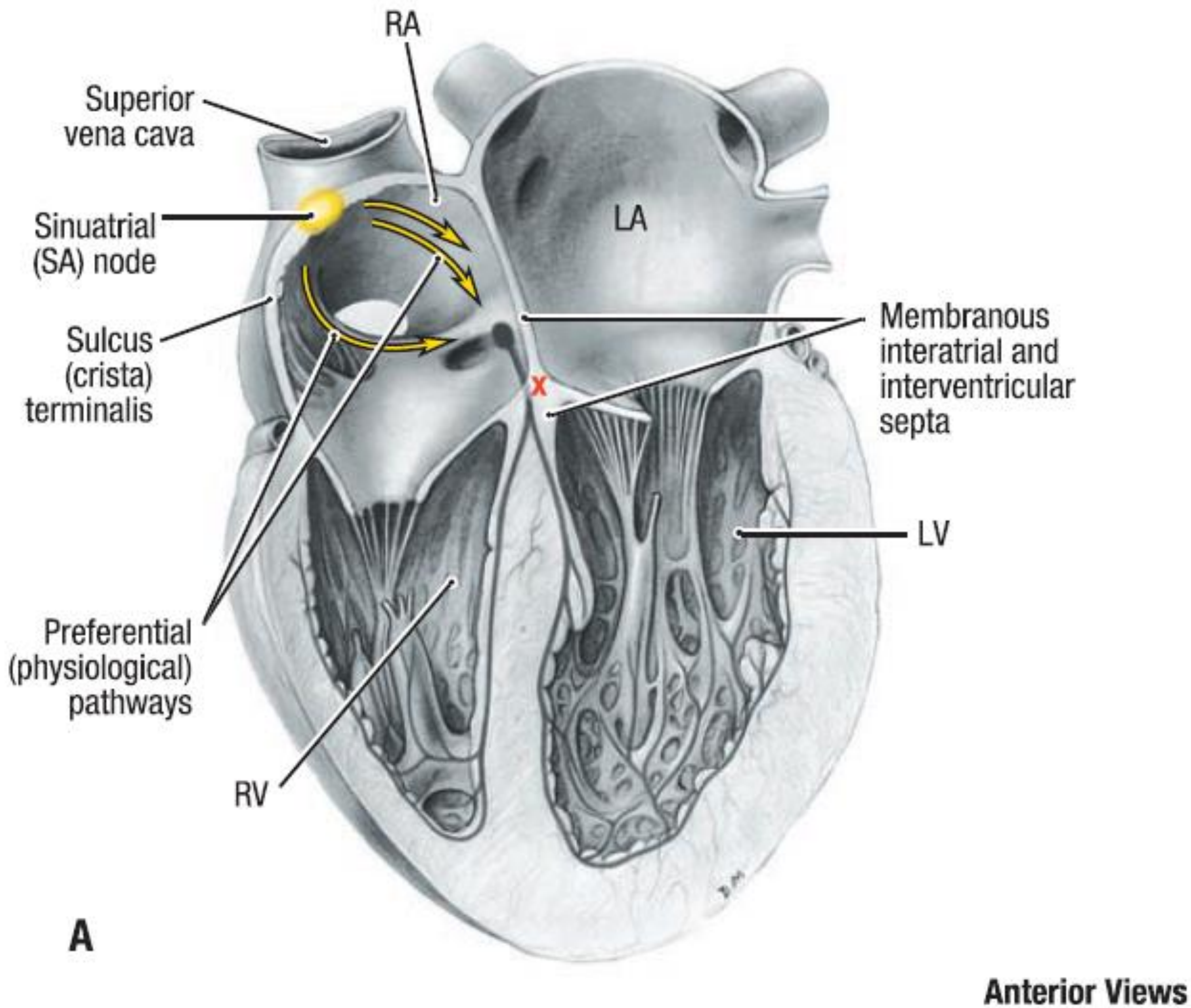


Specialized for initiating impulses and conducting them rapidly through the heart

↓

coordinate contraction of the four chambers

Conducting system of the heart



Sinoatrial node (SA node): **in the wall of the right atrium (*natural pacemaker*)**

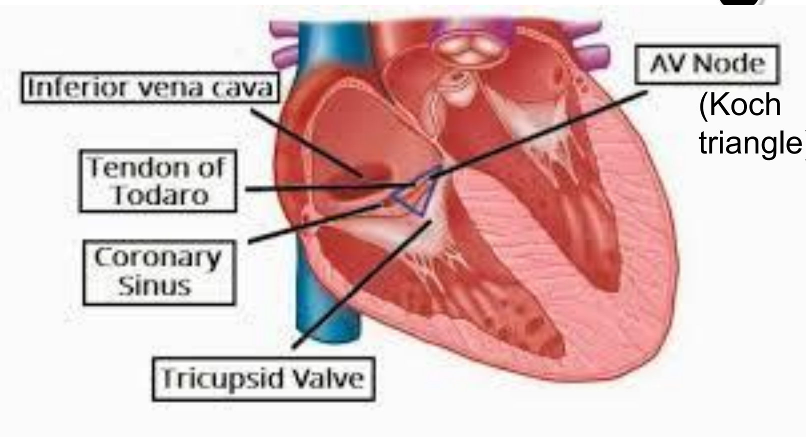
↓

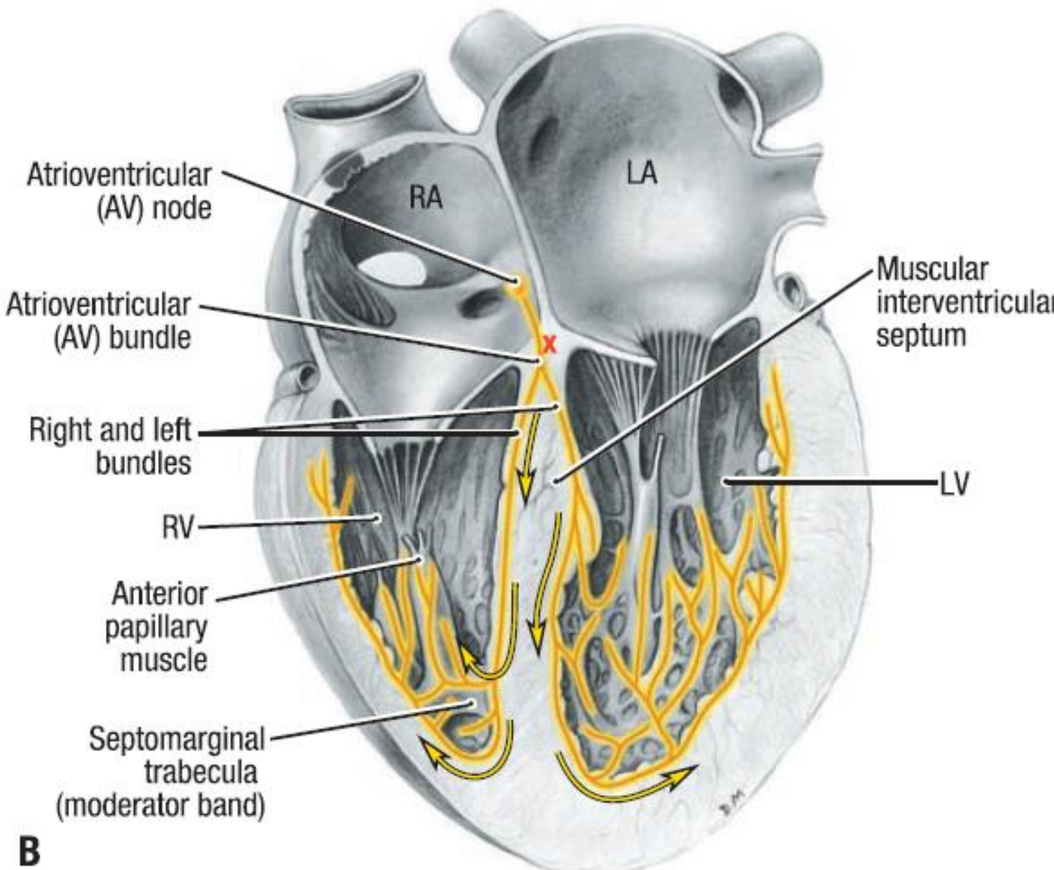
at the superior end of the crista terminalis, at the junction of the sup. v. cava and the right auricle

↓

70 times/ min

Conducting system of the heart





Atrioventricular node: (AV node): **in the interatrial septum, on the ventricular side of the orifice of the coronary sinus**

40 times/min

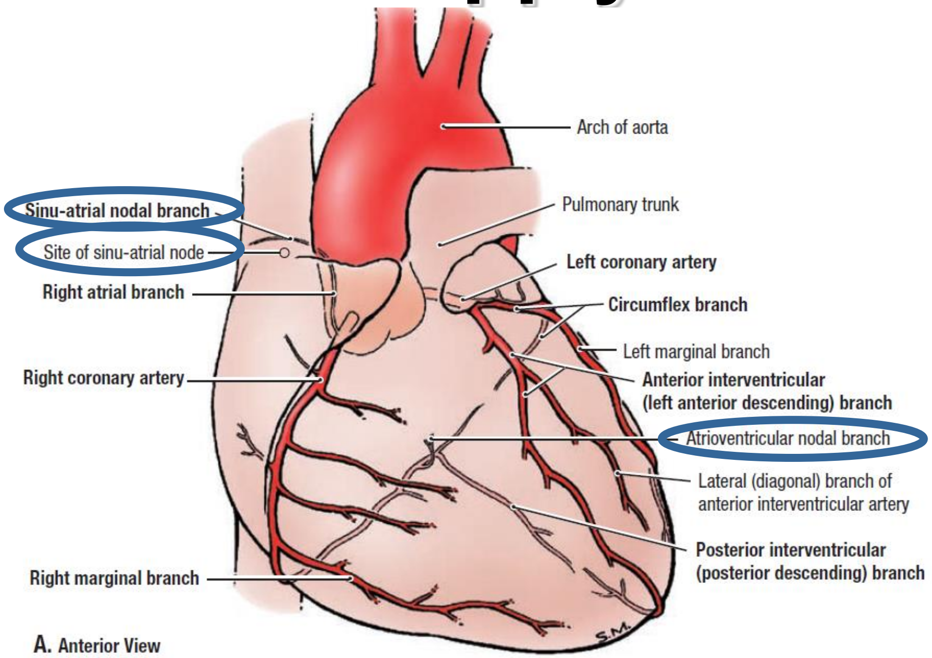
Atrioventricular bundles:

a.) His bundle: **anulus fibrosus (right fibrous trigone)**

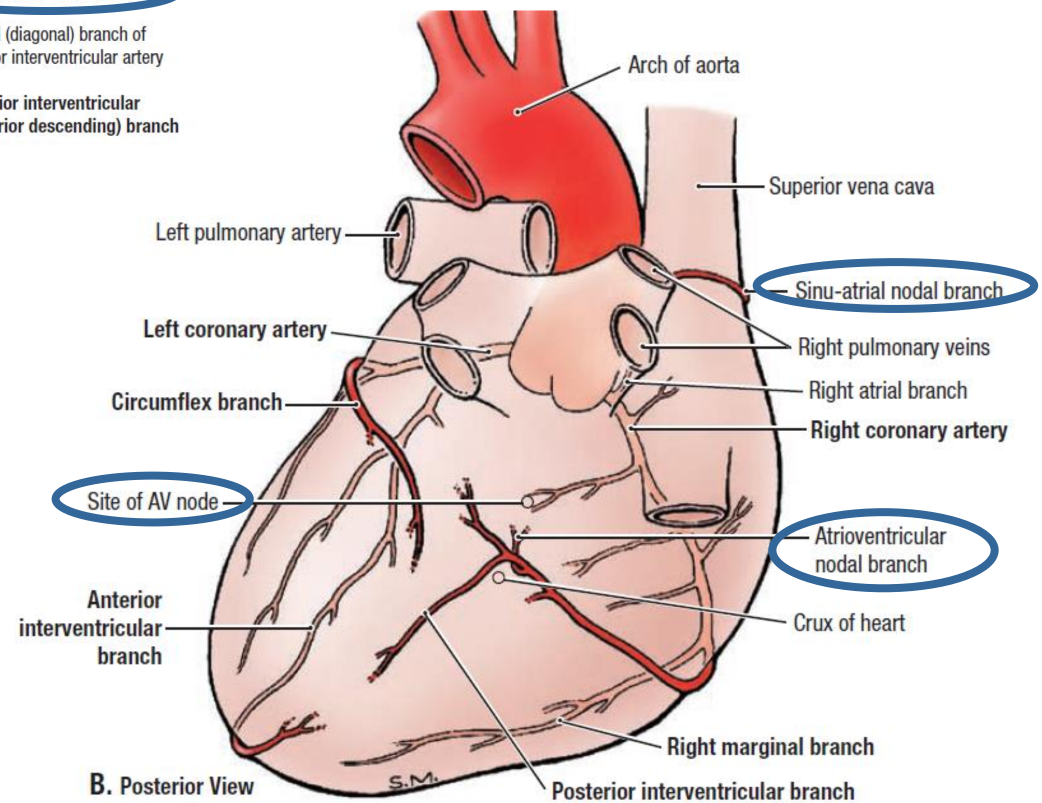
b.) Tawara fibers**: (left-right - subendocardial) in the muscular interventricular septum**

c.) Purkinje fibers **(to the papillary muscles)**

Blood supply of the conducting system



A. Anterior View



B. Posterior View

Innervation of the heart

Innervation of the heart

Autonomic nerve fibers:

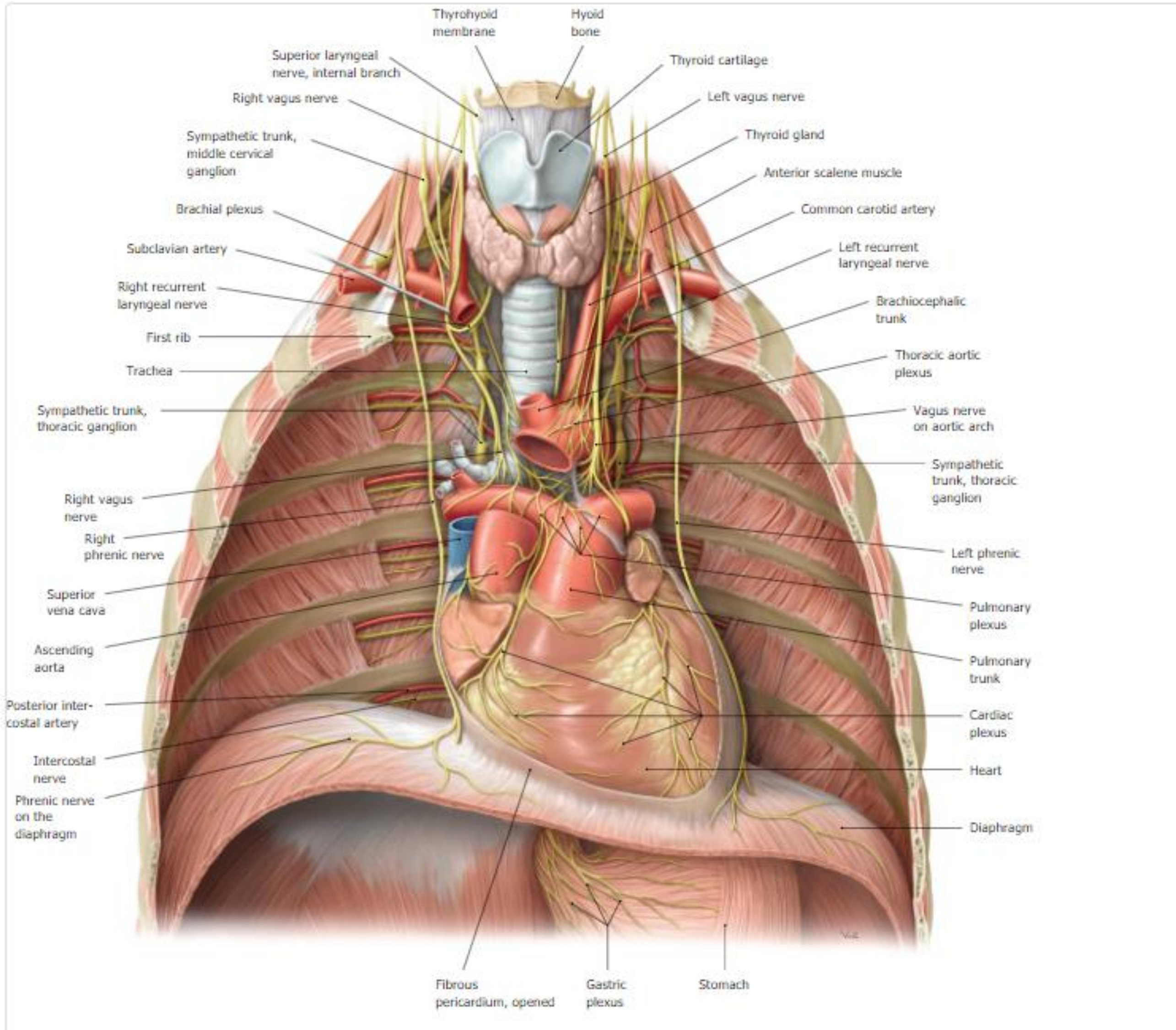
- **sympathetic trunks**
- **vagus nerves**

***Sympathetic stimulation*:**

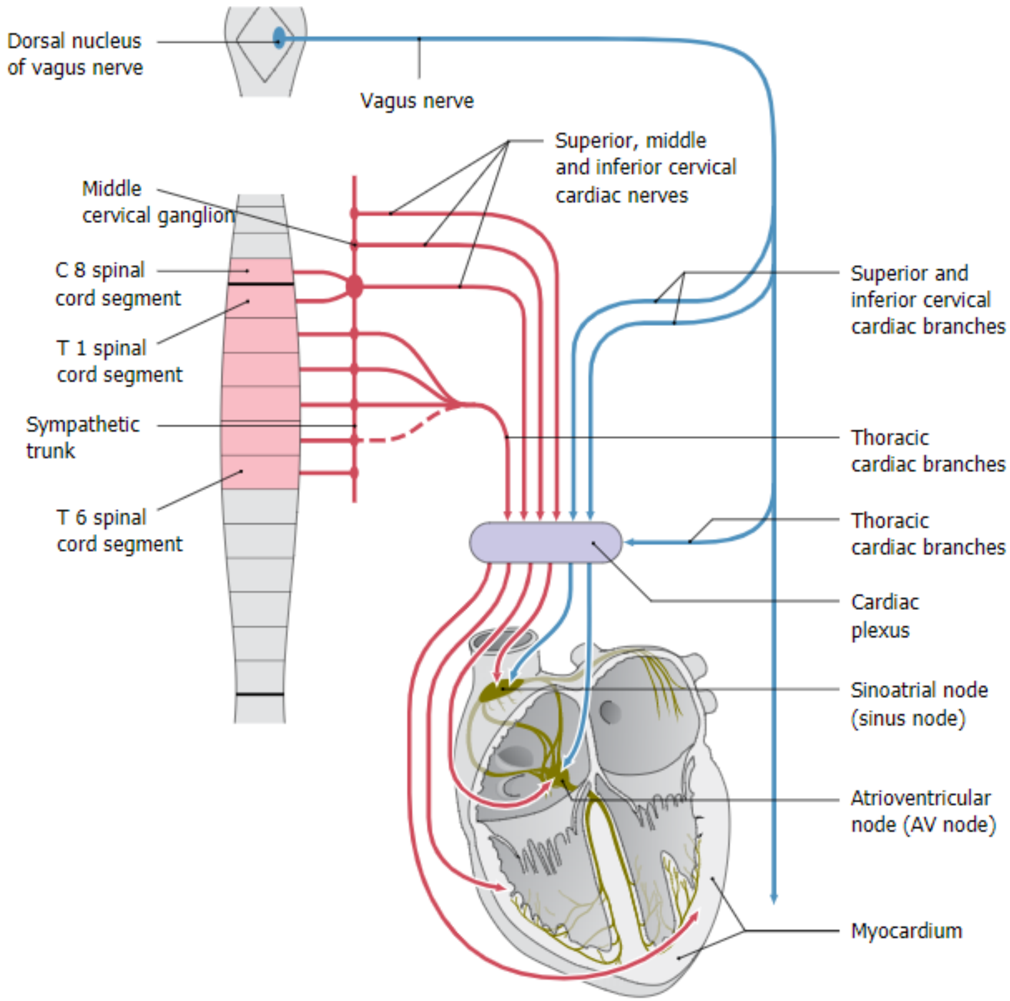
- **increases the heart rate**
- **increases the heart beat**
- **causes dilatation of the coronary arteries**

Parasympathetic stimulation:

- **decreases the heart rate**
- **reduces the heart beat**
- **constricts the coronary arteries**



Innervation of the heart



sympathetic - sympathetic trunk ganglions:
- superior, middle and inferior cervical cardiac branches
- thoracic cardiac branches

parasympathetic - vagus nerve:
- superior and inferior cervical cardiac branches
- thoracic cardiac branches (intramural ganglions)

sensory fibers:
join the
- sympathetic fibers (DRG) or
- vagus nerve (inferior ganglion)